IT Project Management

Lecture 8:
Project **Quality** Management

—Ensuring Excellence in Project Deliverables—

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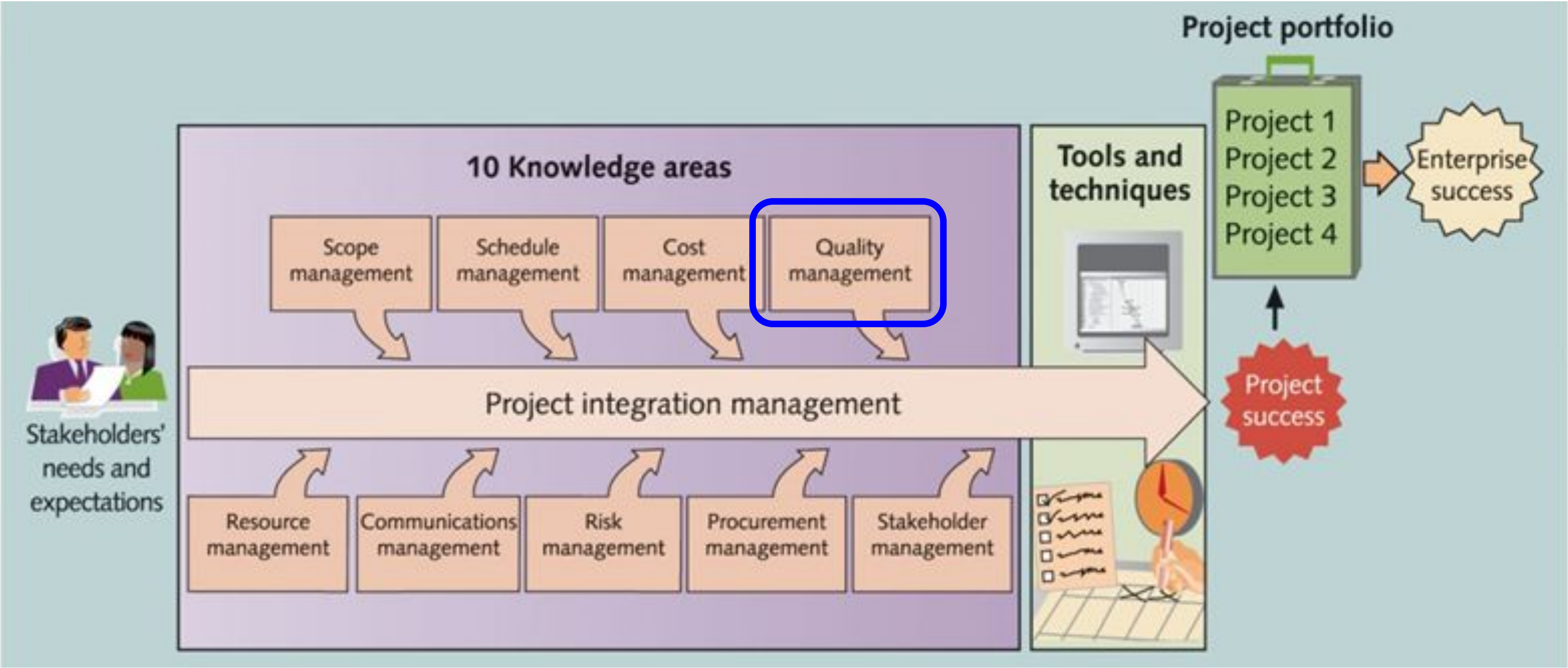
Project **Quality** Management



About this lecture

📌 Learning objectives:

- Define **project quality** and its importance
- Explain the **key processes** in project quality management
- Identify **tools and techniques** for managing quality
- Understand **quality management models** in real-world projects

Introduction to Project Quality Management

What is Quality?

ISO definition (International Organization for Standardization):

- **Quality** is the degree to which a product or service
    - meets **specified requirements (standard)**

General definition:

- **Quality** is the degree to which a product or service
    - is **suitable for use** and satisfies **stakeholder expectations**

→ Focuses on both **performance** and **conformance**

Key Aspects of Quality

- **Performance Quality**: How well the product works (e.g., speed, efficiency)
- **Conformance Quality**: Following standards and specifications (e.g., ISO)
- **Usability**: Ease of use and user experience
- **Reliability**: Consistency over time without defects

Quality in IT: **Example**

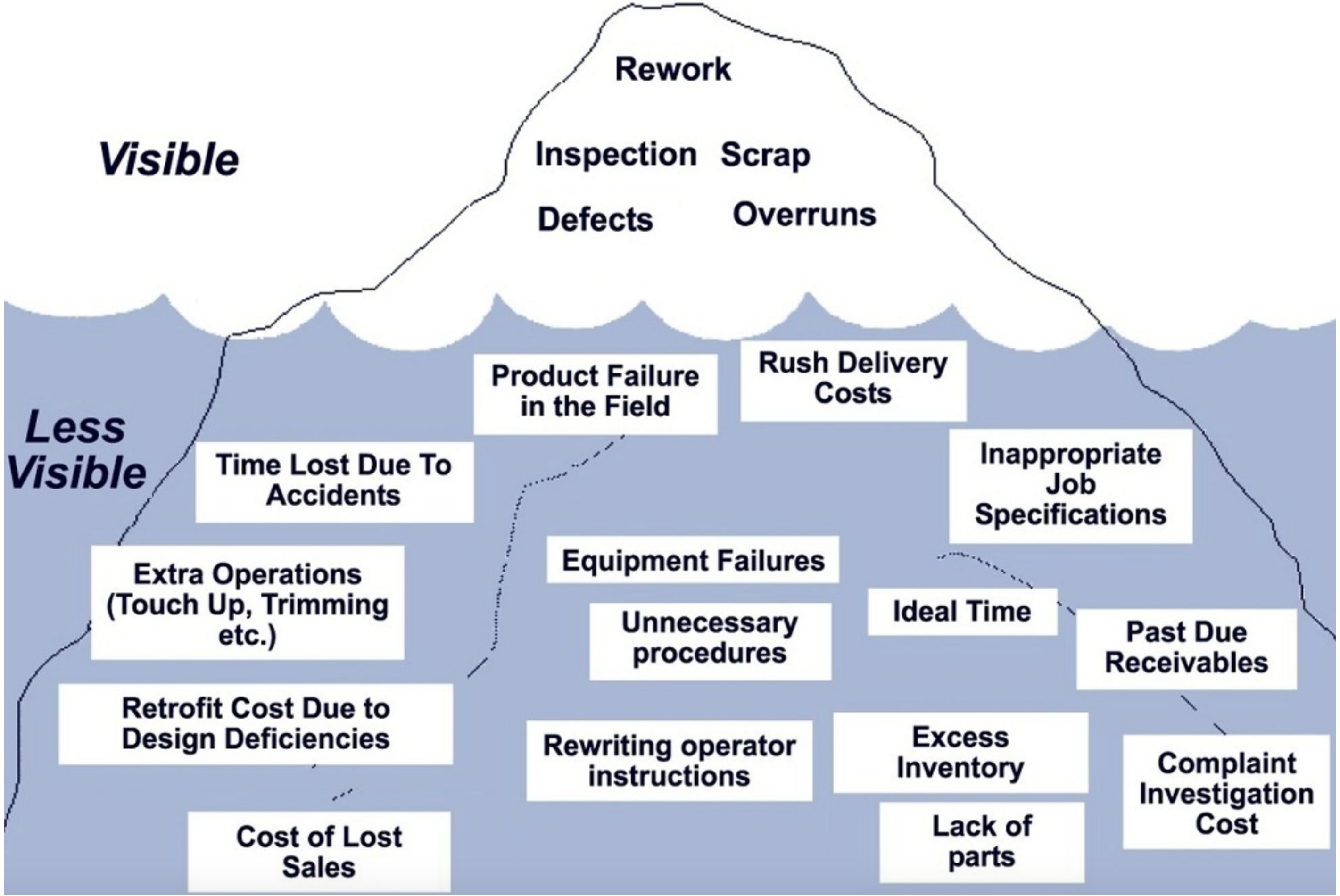
A **mobile banking app** **works well** and has **advanced features**. However, it **crashes frequently** and has **security issues**.

Question: Do you think this app has high quality?

Performance Quality: High (fast and feature-rich)
Conformance Quality: Low (unstable, not secured)

→ **Low quality overall**

Cost of Poor Quality (COPQ)



Cost of Poor Quality: 4 Categories

1. **Prevention Costs**: Training, process improvements
2. **Appraisal Costs**: Inspections, testing
3. **Internal Failure Costs**: Rework, scrap materials
4. **External Failure Costs**: Warranty claims, lawsuits, reputational damage

Poor Quality Control: **Example**

Case: Toyota's Recall Crisis (2009–2010)

Faulty accelerator pedals led to **millions of car recalls** worldwide

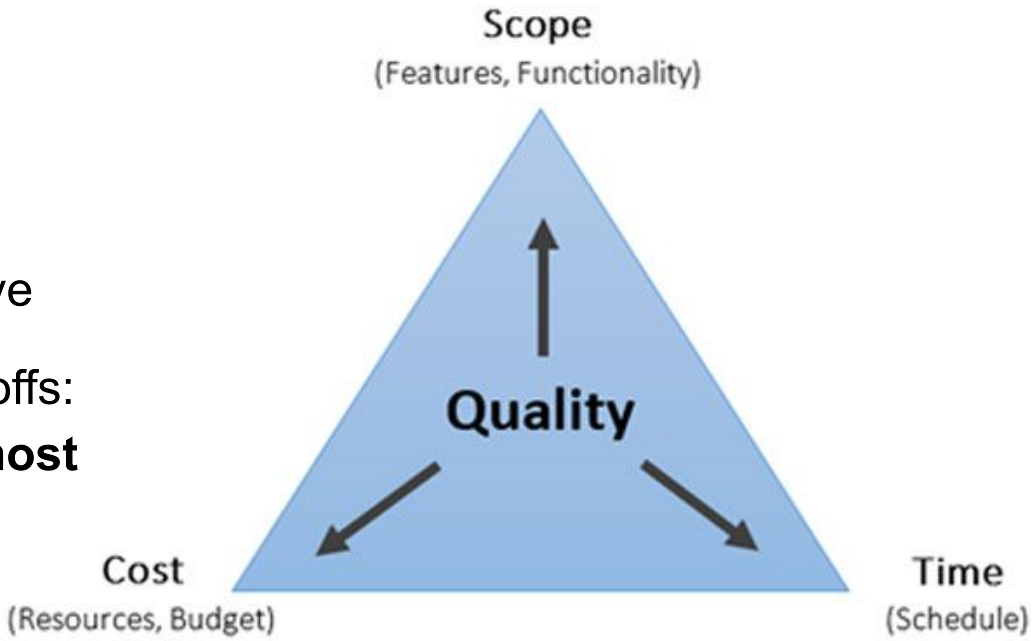
→ Poor quality control resulted in **customer injuries, lawsuits, and lost trust**

→ **Cost of poor quality:** Over **$2 billion in recall expenses**

Balancing Quality with Scope, Time, and Cost

- Fast + Cheap = **Low Quality**
- Cheap + **High Quality** = Slow
- Fast + **High Quality** = Expensive

→ Quality requires balancing trade-offs: you must **prioritize what matters most** for your project



How Project Quality Management Can Help

- Reduces **Defects and Rework**
- Saves **Time and Costs**
- Enhances **Customer Satisfaction**
- Ensures Compliance with **Regulations**

Project Quality Management Processes

The 3 Key Steps of Project Quality Management Process

1. **Plan Quality Management:** Setting Standards
2. **Perform Quality Assurance (QA):** Ensuring Compliance
3. **Perform Quality Control (QC):** Monitoring & Inspecting

Step 1: Plan Quality Management

- Identify **industry standards and best practices** & define **quality objectives, requirements, and success criteria**
- Develop a **Quality Management Plan** for the project
- Define **Quality Tools & Techniques** to be used in the project

Step 1: Plan Quality Management

Example:

Identify key quality **metrics** for a **mobile app development project**

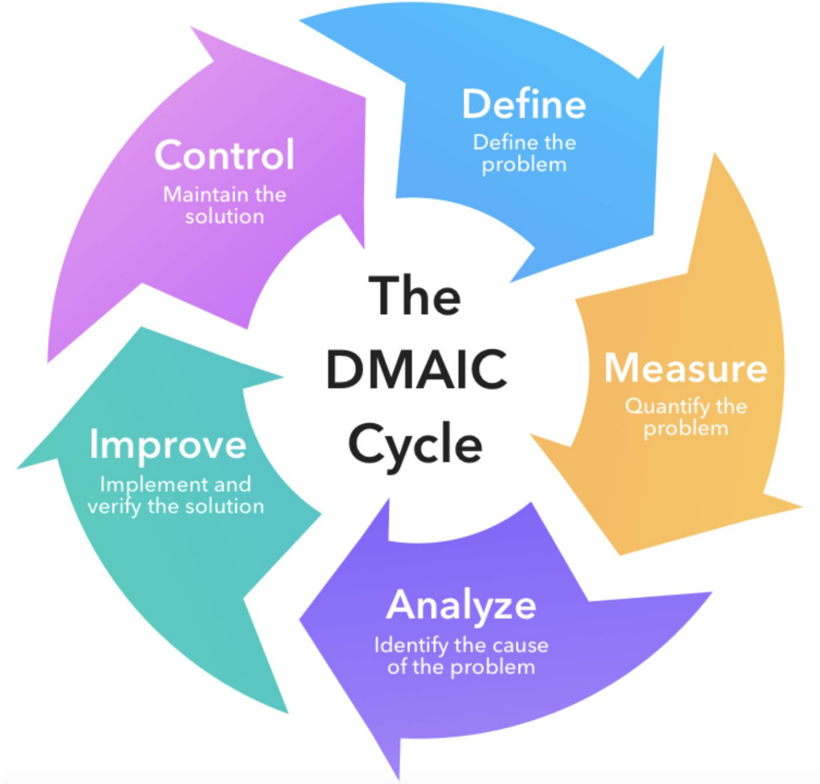
Category	Metric	Target
Performance	Load Time	<2 sec
	Response Time	<500ms
Usability	Task Success Rate	>90%
Reliability	Crash Rate	<1 per 10,000 sessions
Security	Critical Vulnerabilities	0
User Satisfaction	App Store Rating	>4.5 stars

Step 2: Perform Quality Assurance

- **Proactive process:** Proactively preventing defects in project deliverables
- Uses **process audits, peer reviews, and best practice evaluations**

Step 2: Perform Quality Assurance

Example:
A manufacturing company implements **Six Sigma** to reduce defects in production



Step 3: Perform Quality Control

- **Reactive process:** Reactively detecting defects in project deliverables
- Involves **testing, inspections, and defect tracking**

Note: Defect prevention (QA) is often **cheaper** than **defect correction (QC)**

Step 3: Perform Quality Control

Example:

A software team uses **bug tracking tools** to analyze defect trends

Key Tools in Quality Management

Key Tools in Quality Management

1. Cause-and-Effect Diagram (Ishikawa/Fishbone Diagram)
2. Check Sheets
3. Pareto Charts
4. Histograms
5. Control Charts
6. Scatter Diagrams
7. Flowcharts

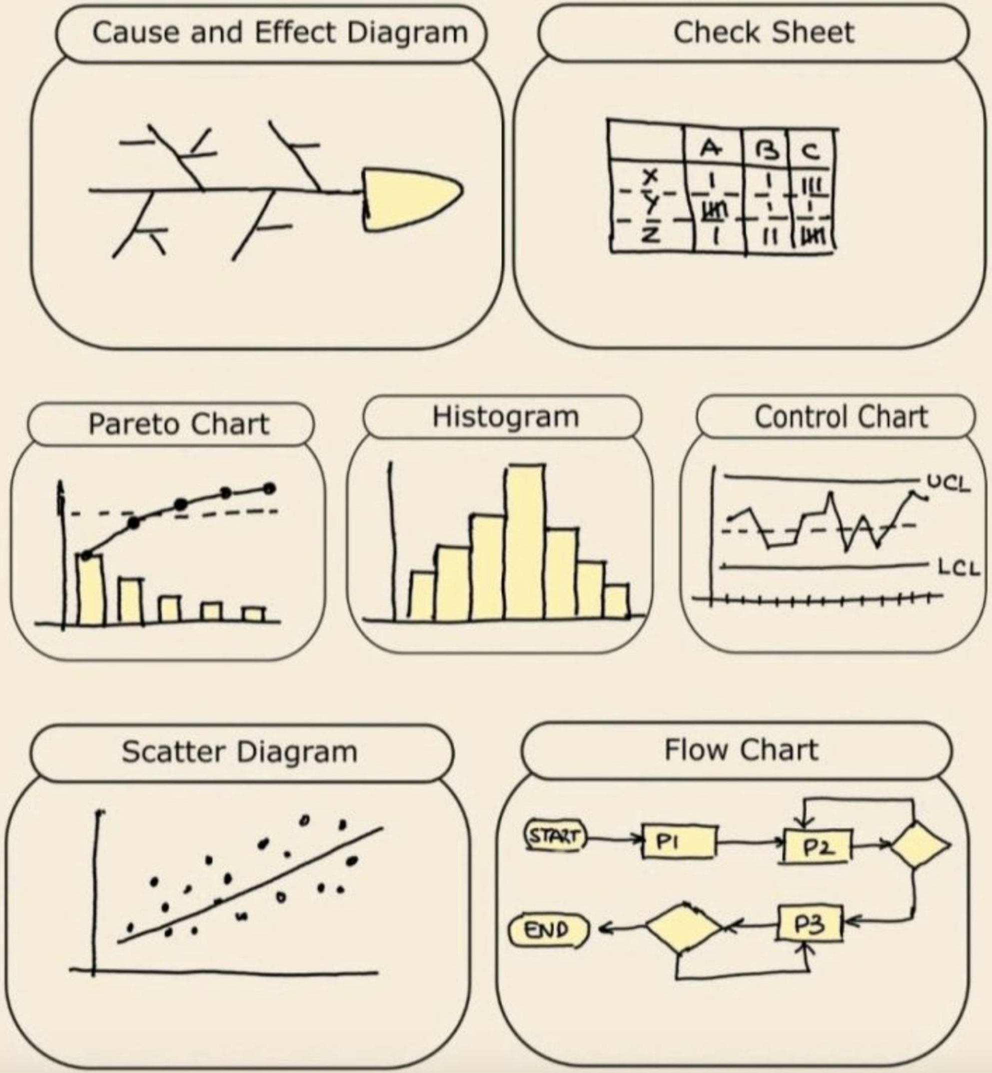
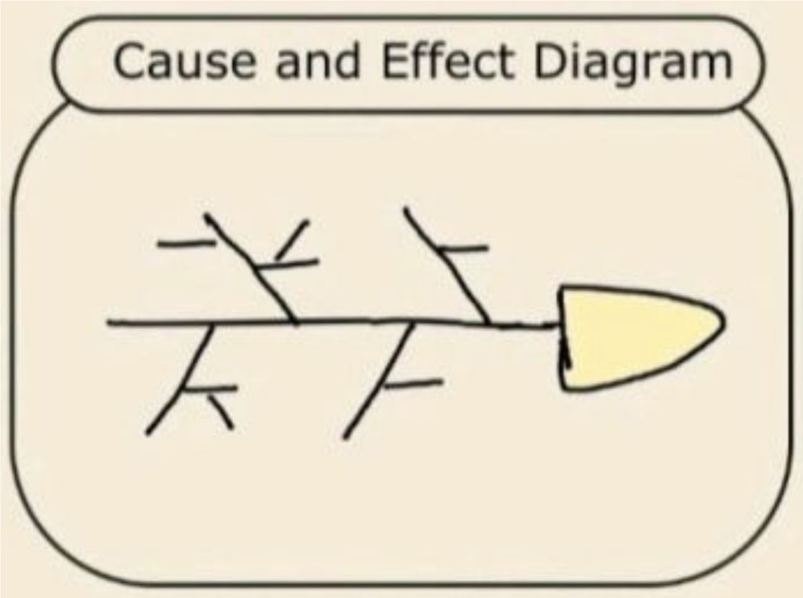


Image: QualityGurus.com

Cause-and-Effect Diagram (Ishikawa/Fishbone)

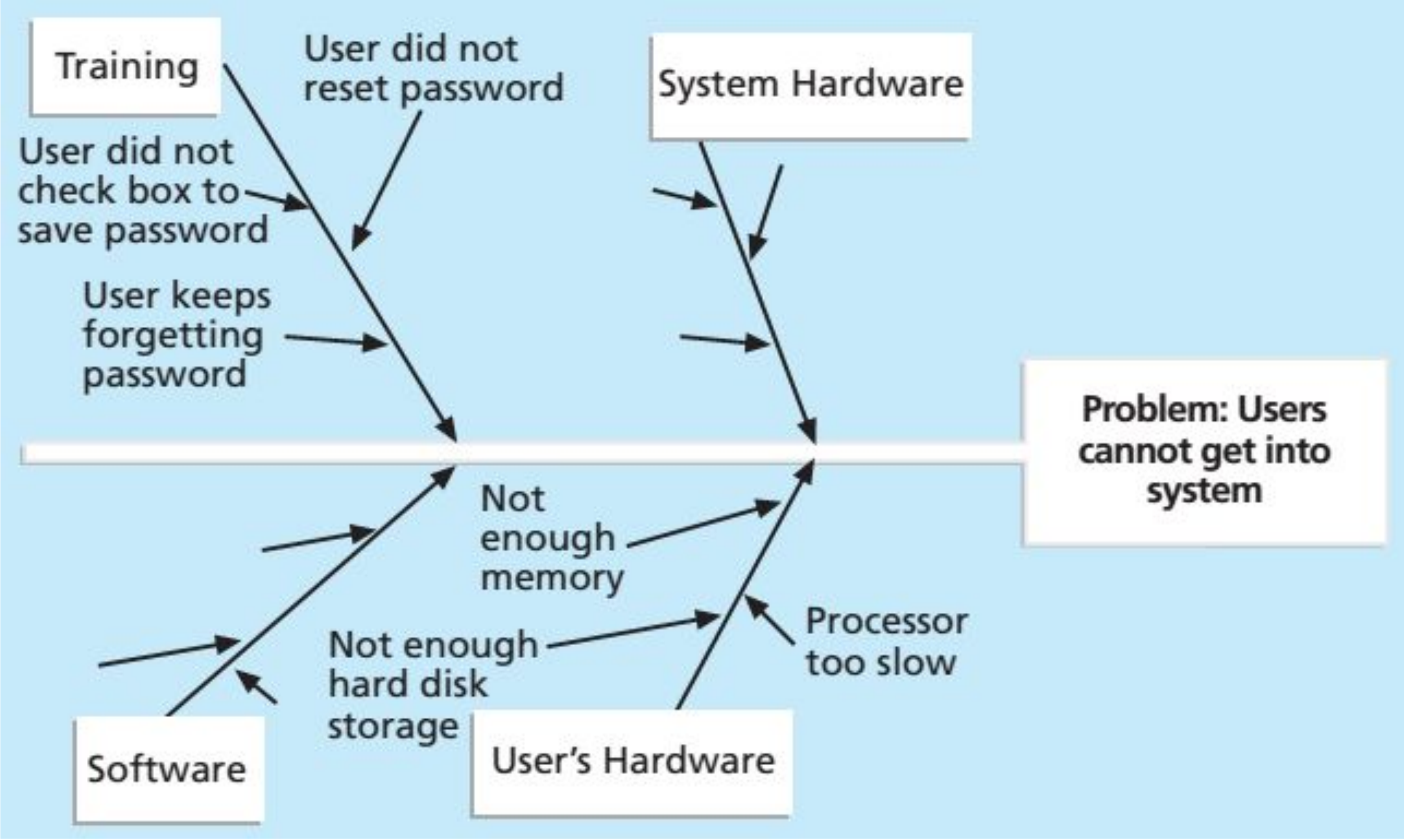
- Identifies **root causes** of a problem
- Helps teams **brainstorm possible reasons** behind defects
- Categories when brainstorming: People, Process, Materials, Equipment, Environment



Cause-and-Effect Diagram (Ishikawa/Fishbone): **example**

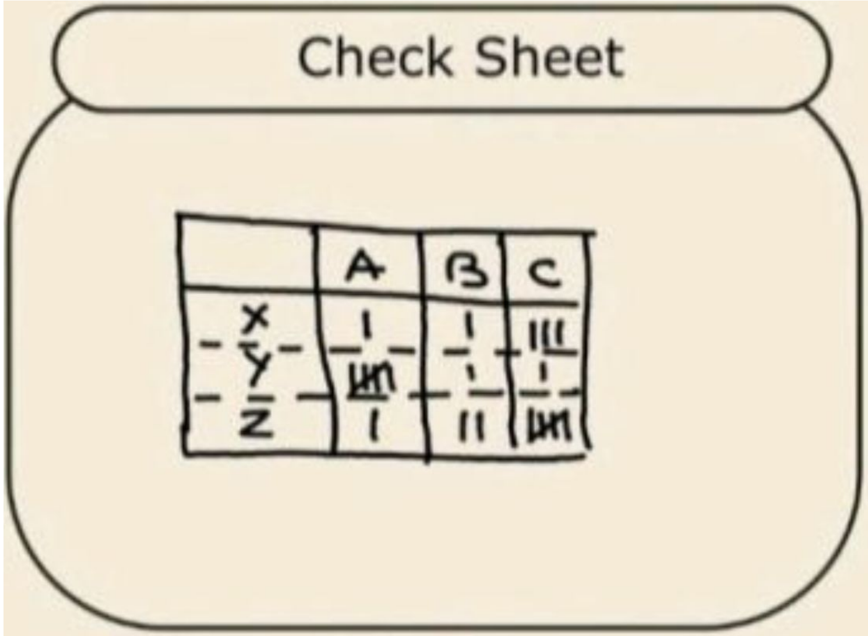
User reports a vague bug

→ The team uses fishbone diagram to analyze and find the root causes



Check Sheets

- Simple **data collection** tool used for tracking defects, errors, or issues
- Standardized format ensures consistency
- Helps in analyzing **patterns and frequency** of defects



Check Sheets: **example**

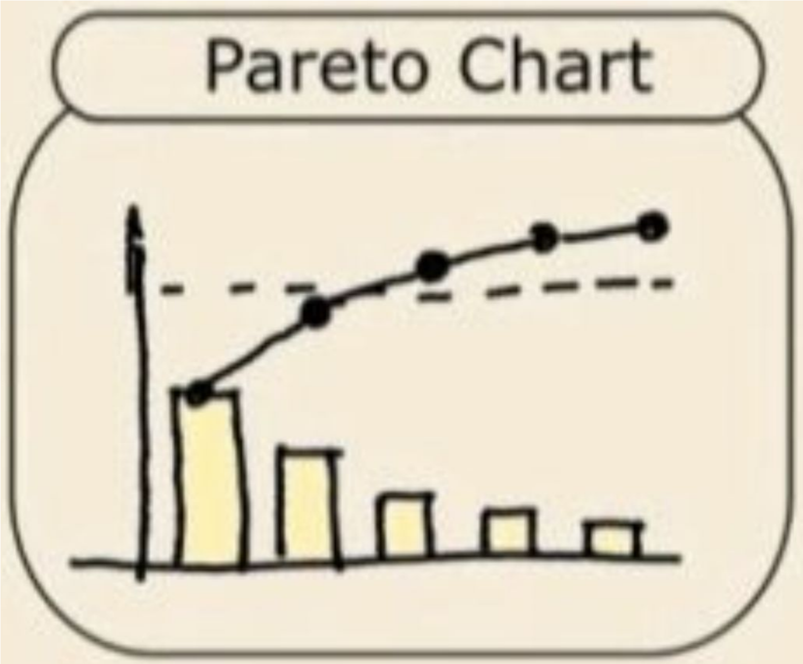
IT support team records the common patterns of user complaints

System Complaints

Source	Day							
	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Total
Email	𝍷𝍷𝍷	𝍷𝍷𝍷𝍷	𝍷	𝍷𝍷	𝍷		𝍷	12
Text	𝍸 𝍷𝍷	𝍷𝍷𝍷𝍷	𝍸 𝍷	𝍷𝍷𝍷	𝍷𝍷𝍷𝍷	𝍷𝍷	𝍷𝍷𝍷	29
Phone call	𝍷	𝍷𝍷	𝍷	𝍷	𝍷𝍷	𝍷		8
Total	11	10	8	6	7	3	4	49

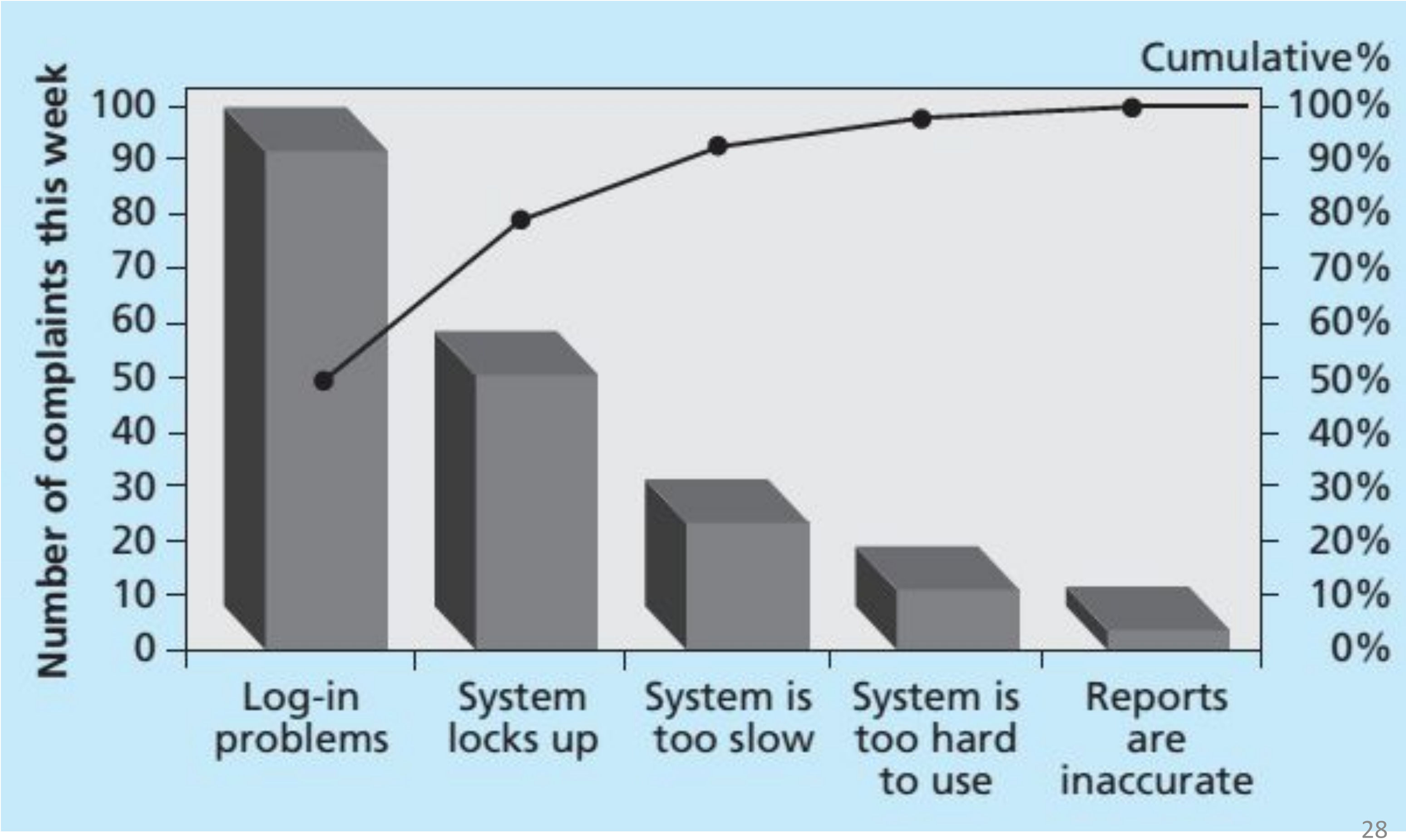
Pareto Charts (80/20 Rule)

- Helps prioritize the **most significant issues** using the **Pareto Principle**
- Focuses on solving **20% of causes** that lead to **80% of problems**
- Combines **bar graph (individual impact)** and **line graph (cumulative impact)**



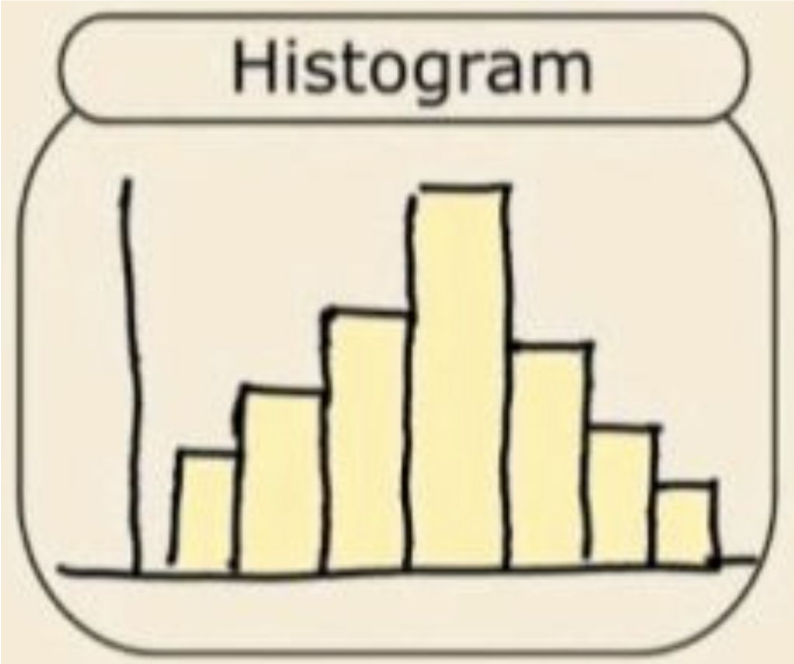
Pareto Charts (80/20 Rule): **example**

A Pareto chart shows that 20% of bugs cause 80% of user complaints



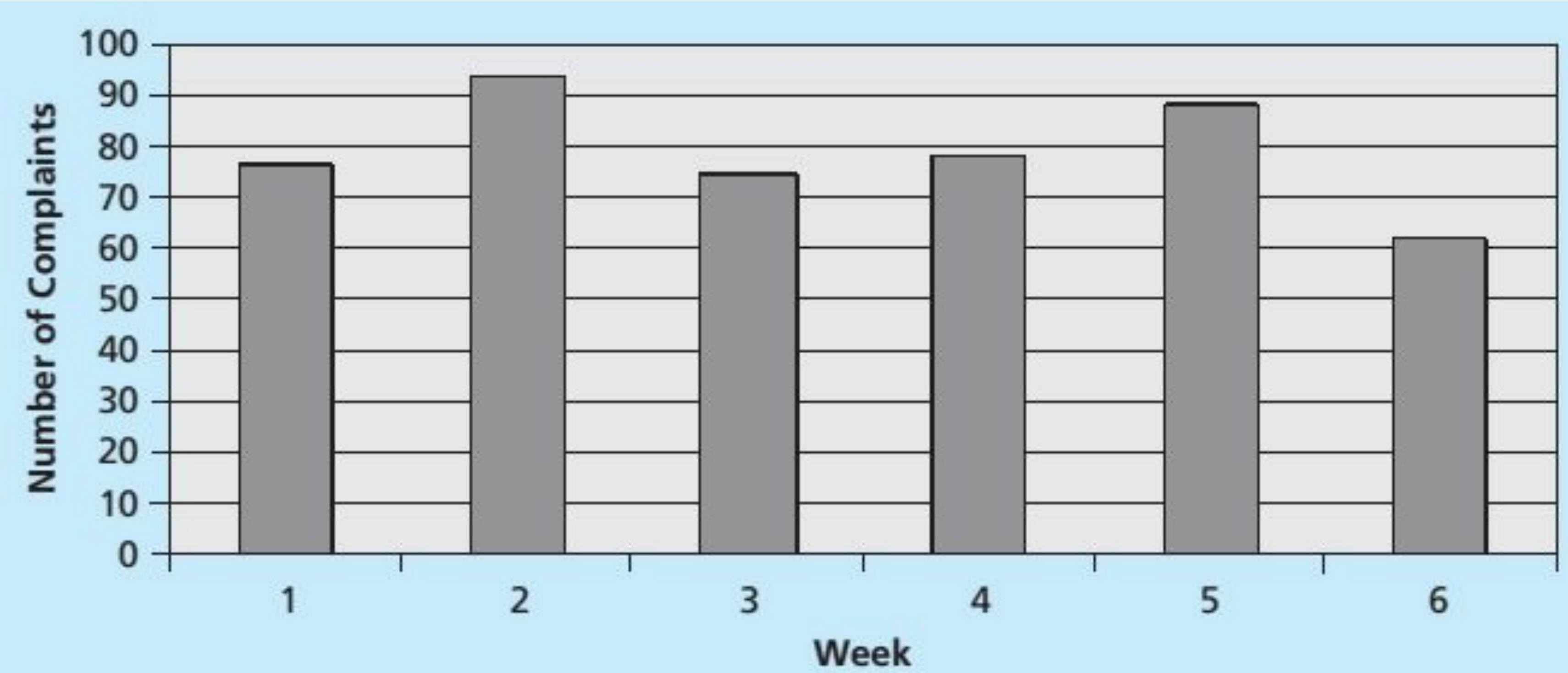
Histograms

- Bar chart representation of **frequency or count**
- Shows data **distribution, variations, and trends**



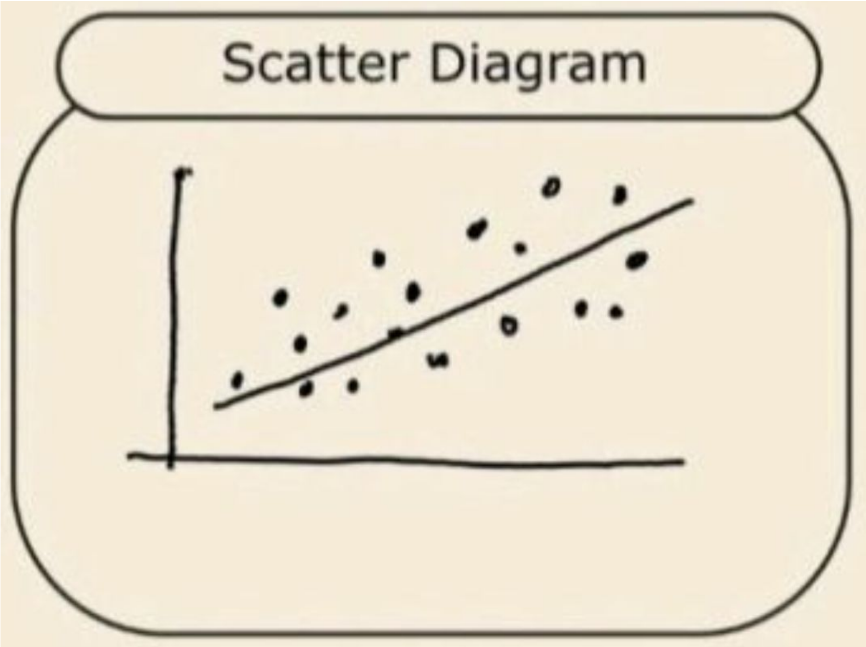
Histograms: **example**

A histogram show the number of complaints per release week

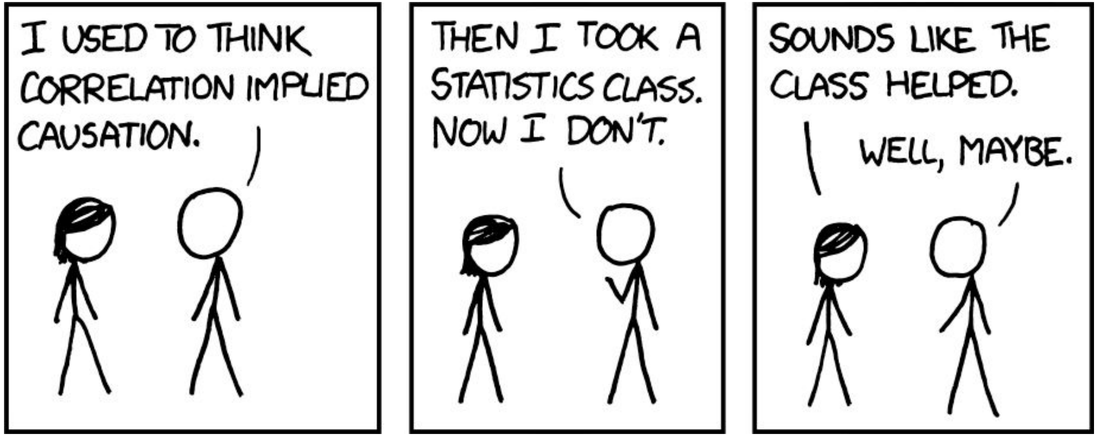


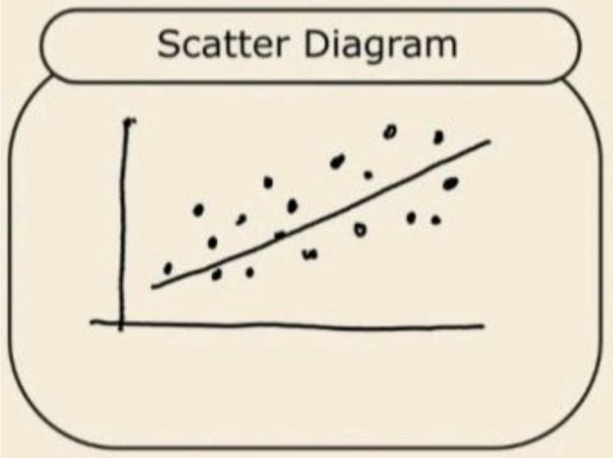
Scatter Diagrams

- Shows **correlation** between two variables
- Identifies **positive, negative, or no correlation** trends



Scatter Diagrams

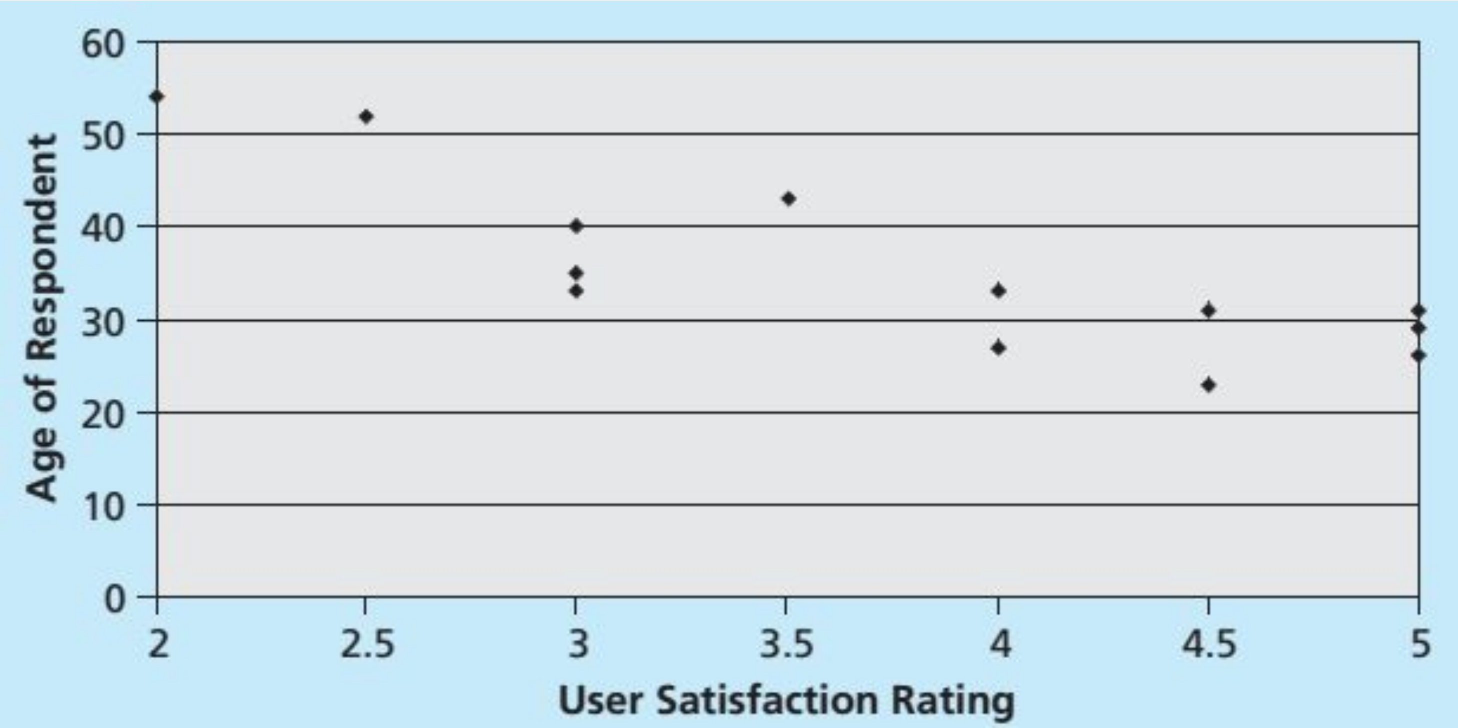




Note: **Correlation** does **not imply causation**, but it may suggest where to look at

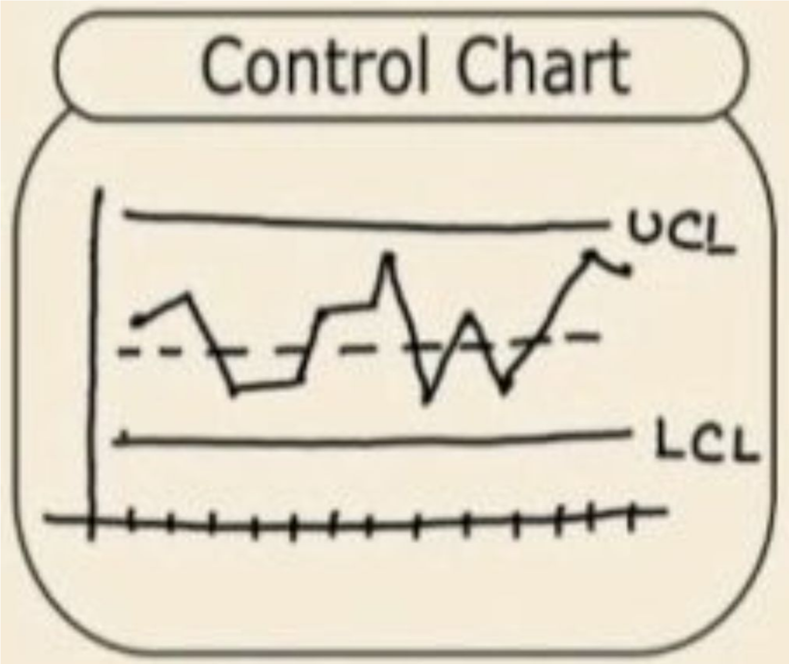
Scatter Diagrams: **example**

A scatter plot shows correlation between users' age and their ratings



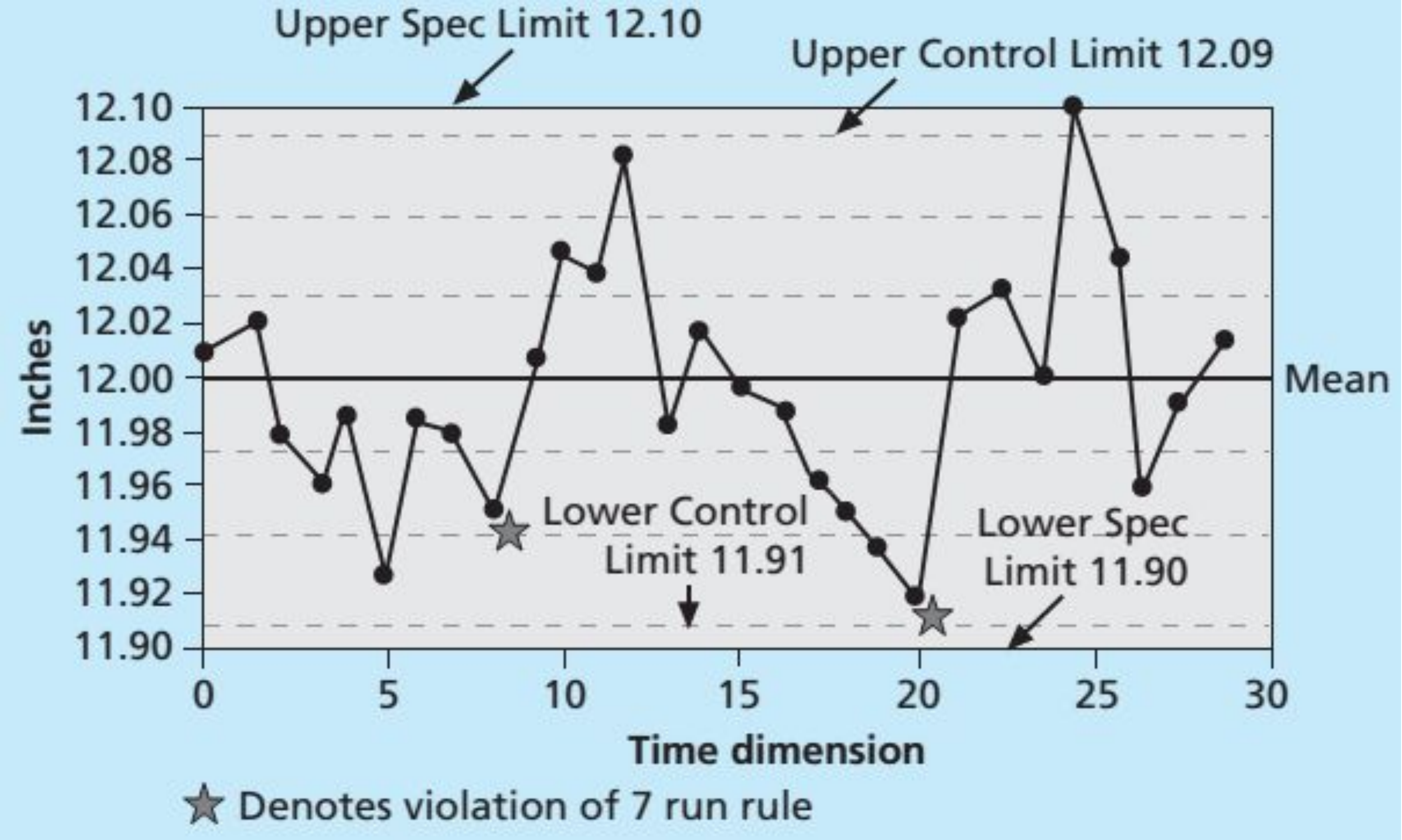
Control Charts

- Monitors process **stability over time**
- Uses **Upper Control Limit (UCL)** and **Lower Control Limit (LCL)** to detect variations
- Helps determine if a process is stable or needs corrective action



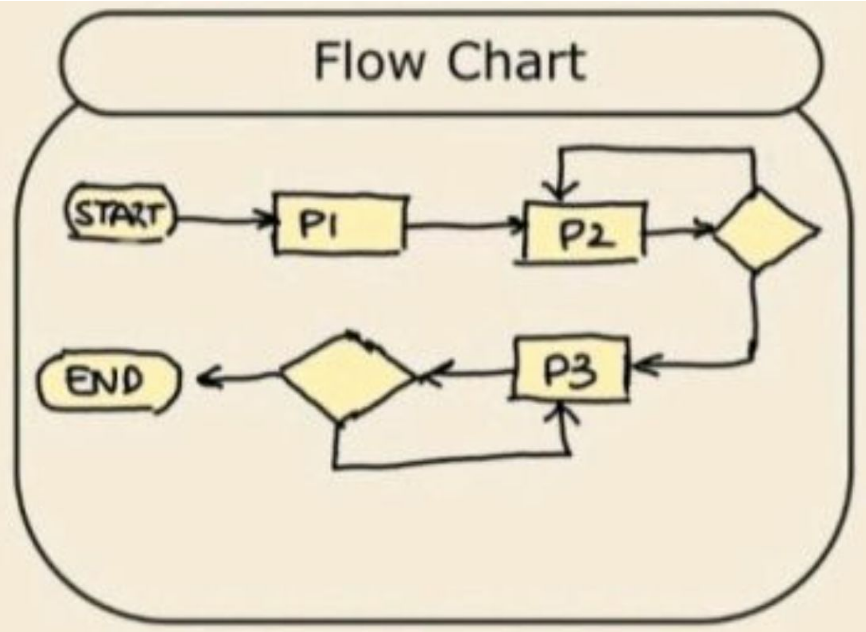
Control Charts: **example**

Manufacturing plants track product defects per day to spot trends



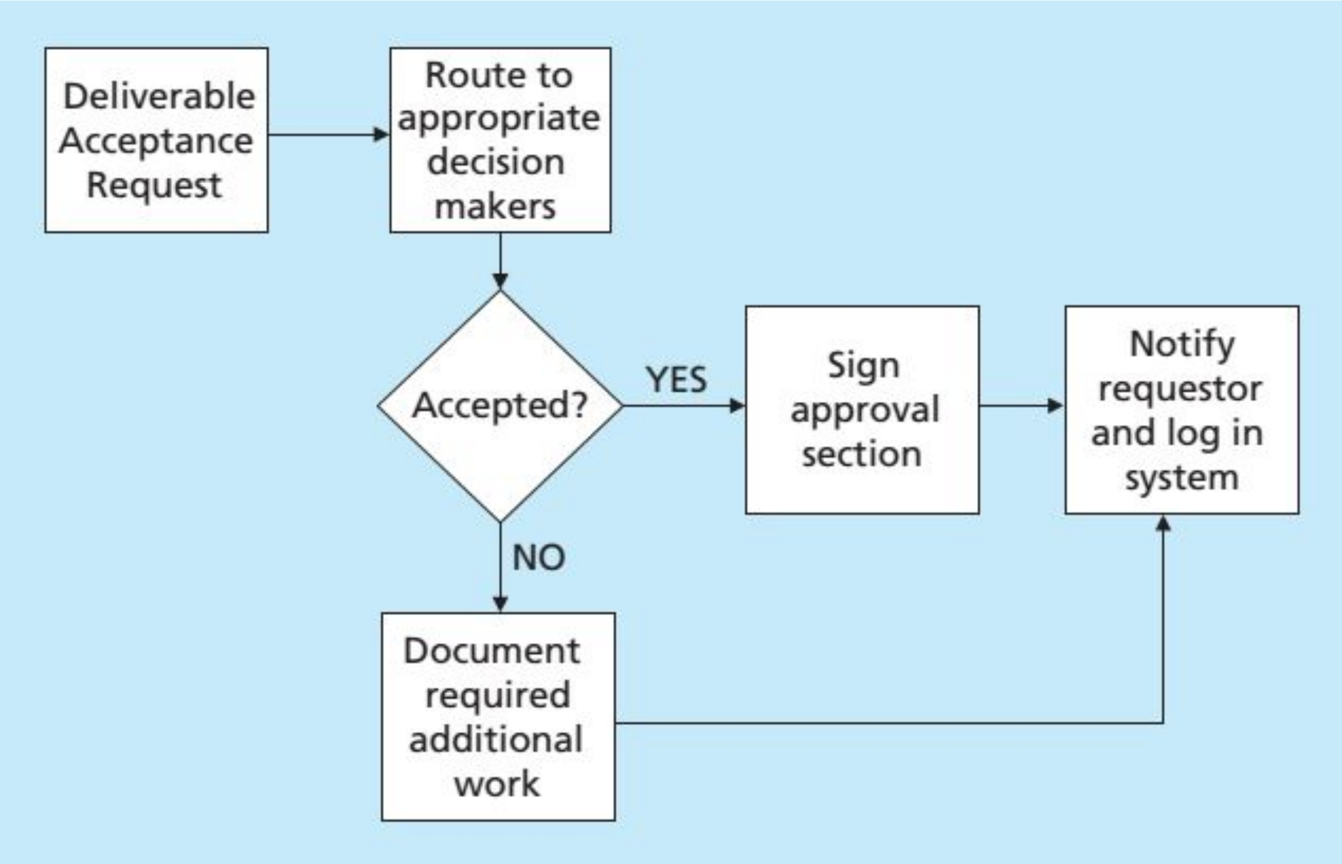
Flowcharts

- Visual representation of a **process flow**
- Helps in **understanding, analyzing, and optimizing workflows**
- Identifies **bottlenecks and inefficiencies** in processes



Flowcharts: **example**

A flowchart visualizes the steps and identifies where delays may occur in a workflow



Quality Management Models

Quality Management Models Overview

- **Deming's 14 Points:** Continuous improvement culture
- **Juran's Quality Trilogy:** Plan, Control, Improve
- **Crosby's Zero Defects:** Prevention over inspection

→ Many industries (IT, manufacturing, healthcare) adopt these models

Deming’s 14 Points

- Focus on **systematic changes** rather than just fixing errors
- Emphasizes **leadership commitment, training, and employee involvement**

Key Concepts:

- Drive out fear: Create a safe environment for innovation
- Quality is **built into the process**, not inspected later
- Improve constantly: Adopt a ***continuous improvement*** mindset

Example:

Toyota’s “**Kaizen** continuous improvements” follow Deming’s principles

Juran's Quality Trilogy

- **Quality Planning:** Identify customer needs and set quality goals
- **Quality Control:** Monitor performance and reduce variation
- **Quality Improvement:** Make continuous improvements over time

Key Idea: Quality does not happen by accident; it must be planned!

Example in software development:

- Quality Planning → Define coding standards
- Quality Control → Perform code reviews & testing
- Quality Improvement → Analyze defects and refine processes

Crosby’s Zero Defects

- **Preventing defects rather than fixing them**
- **Quality is free:** Investing in prevention **saves costs later**

Key Principles:

- **Conformance to Requirements**: Clearly define what quality means
- **Prevention Over Inspection**: Build quality into the process
- **Zero Defects Mentality**: No tolerance for avoidable mistakes

Example:

Airbus ensures zero defects in aircraft parts because small errors = huge risks

Aspect	Deming’s 14 Points	Juran’s Quality Trilogy	Crosby’s Zero Defects
Main Focus	Continuous improvement & leadership	Planning, control, and improvement of quality	Prevention over inspection
Philosophy	Quality is achieved through system improvements	Quality must be planned, controlled, and improved	Zero defects saves money in the long run
Key Concepts	Systematic changes, employee involvement, statistical process control (SPC)	Quality planning, quality control, quality improvement	Conformance to requirements, doing it right the first time
Approach to Defects	Reduce variation & defects through process improvements	Control defects by planning quality into processes	Prevent defects before they happen
Cost of Quality	Focuses on long-term quality improvements that reduce costs over time	Emphasizes cost reduction through structured quality management	Believes quality is free if done right from the start, especially for large projects
Key Quote	"Quality is everyone’s responsibility."	"Quality does not happen by accident; it must be planned."	"Do it right the first time."

Quality Standards & Certifications

Quality Standards & Certifications Overview

- **ISO 9000:** General quality management system
- **CMMI:** Software process improvement
- **Six Sigma:** Defect reduction methodology

→ Benefits:

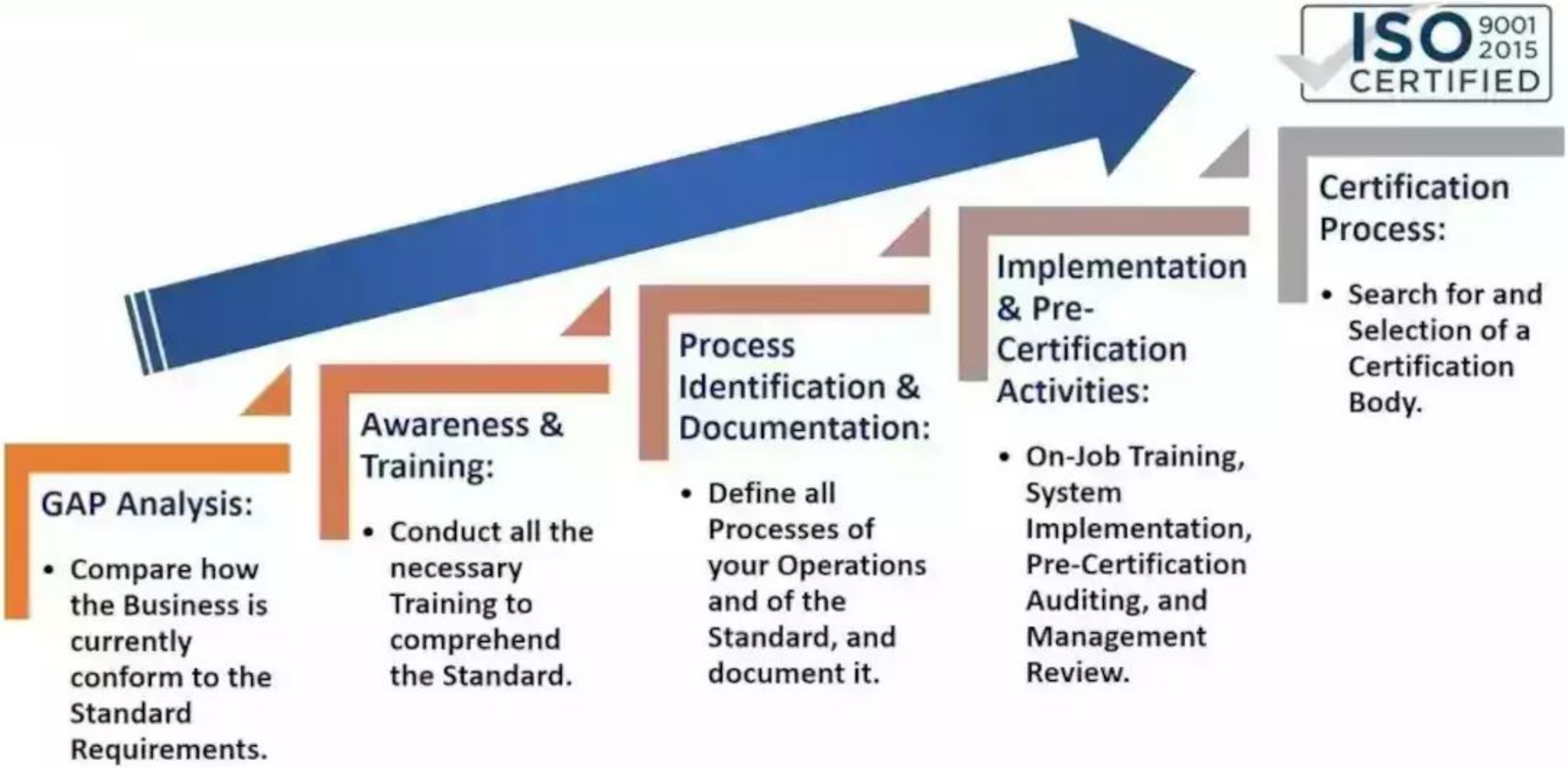
- Provide a structured framework for quality management
- Showcase for improving competitiveness in global markets

ISO 9000: General Quality Management System

- A **global standard** for **quality management systems**
- Ensures **consistent processes, documentation, and customer satisfaction**
- Focuses on a **process-based approach** to quality

Certification: ISO 9001

→ The most widely used standard certification



Steps of ISO 9001 Certification

CMMI (Capability Maturity Model Integration)

- A framework for **evaluating and improving software development processes**
- Focuses on **process maturity levels** from **Initial (chaotic) to Optimizing (best practices)**
- Used in software engineering, defense, healthcare, and banking

CMMI (Capability Maturity Model Integration) – Software Process Improvement

Five Maturity Levels:

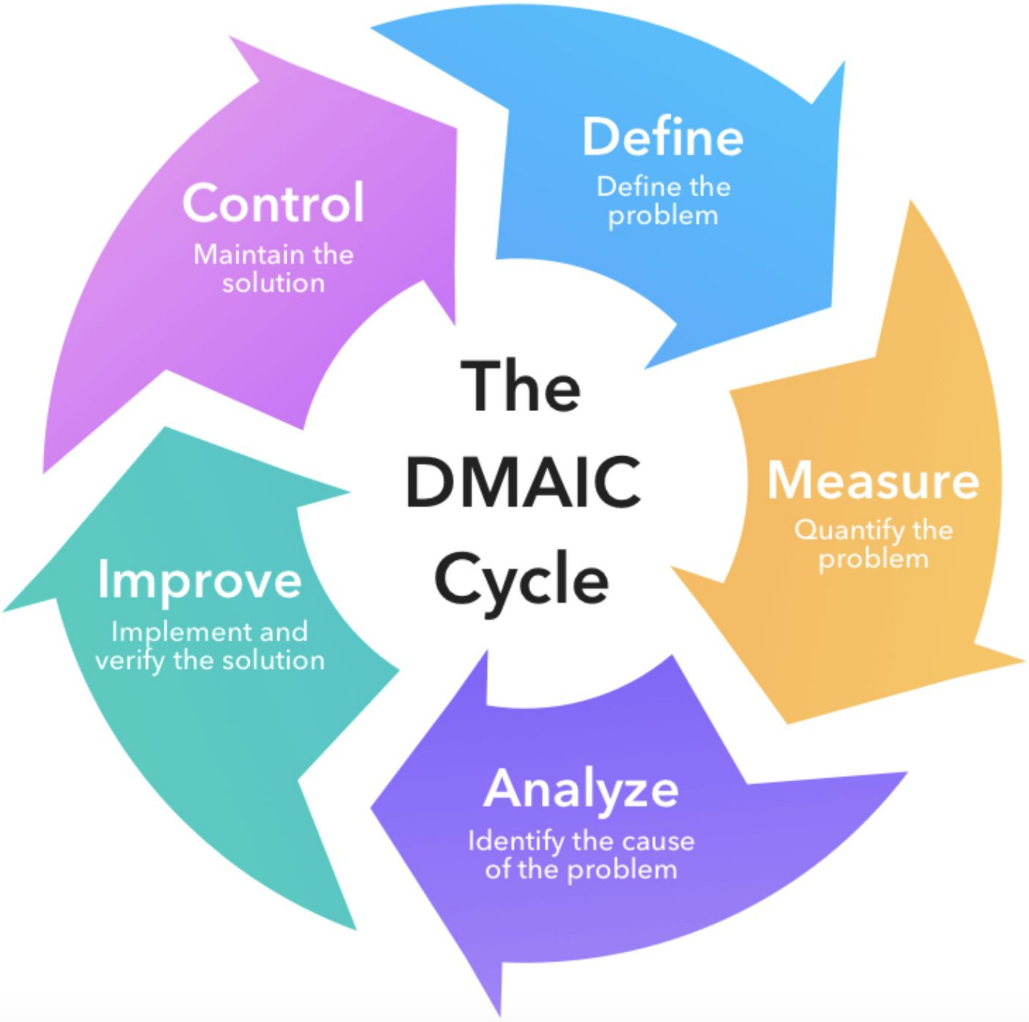
1. **Initial:** Unpredictable processes, reactive management
2. **Managed:** Basic project management processes in place
3. **Defined:** Standardized organization-wide processes
4. **Quantitatively Managed:** Data-driven decision-making
5. **Optimizing:** Continuous process improvement using innovation

Six Sigma

- A methodology focused on **reducing defects** and **improving process efficiency**
- Uses **statistical analysis** to reduce errors to **6-sigma level** (**3.4 defects per million** opportunities)
- Developed by Motorola, later adopted by General Electric (GE), Toyota, and other companies

Six Sigma

Key Concepts: DMAIC Cycle

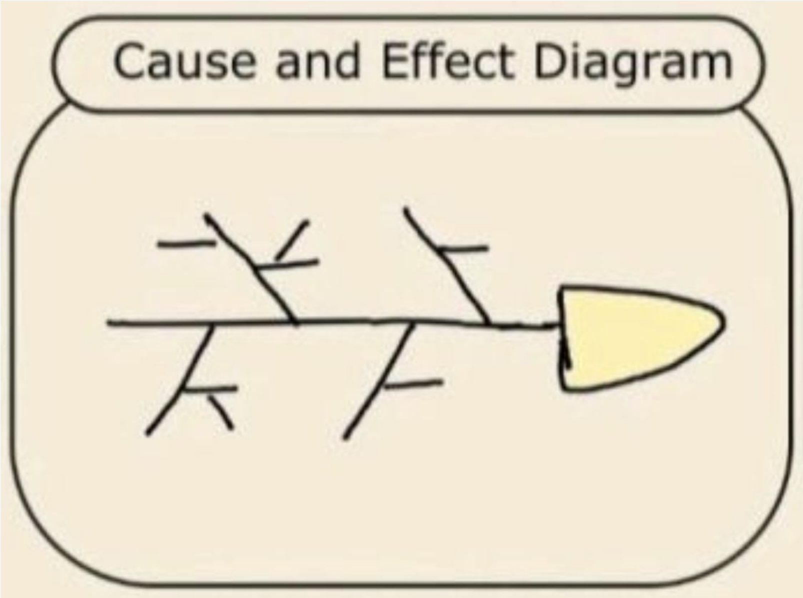


Feature	ISO 9000	CMMI	Six Sigma
Focus	Standardizing quality management processes	Improving software & system processes	Reducing defects & process variation
Industry	All industries	Mainly software & engineering	Manufacturing & business processes
Key Concept	Quality Management System (QMS)	Five maturity levels	DMAIC (Define, Measure, Analyze, Improve, Control)
Goal	Consistency & customer satisfaction	Process efficiency & predictability	Reducing errors to near zero
Certification Levels	ISO 9001	Levels 1-5	Belts (White, Yellow, Green, Black, Master Black)
Benefit	Efficiency & compliance	Better software development	Cost reduction & quality improvement
Weakness	Rigid & requires audits	Focuses on processes, not results	Complex & data-intensive

Exercise

Fishbone Diagram: **Exercise 1**

- Draw a fishbone diagram to identify the **root causes** and **suggest fixes**
- **The issue:** *Students missing assignment deadline*

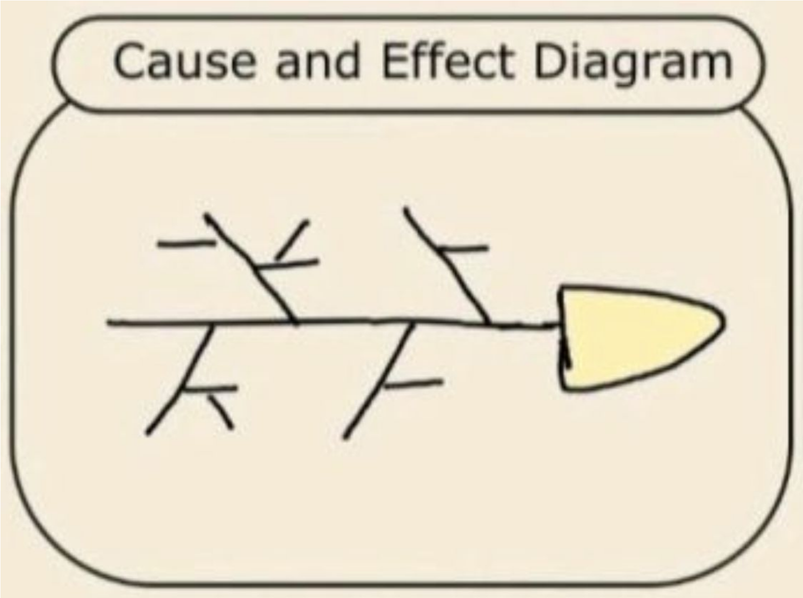


Note:

Categories when brainstorming: People, Process, Materials, Equipment, Environment

Fishbone Diagram: **Exercise 2**

- Draw a fishbone diagram to identify the **root causes** and **suggest fixes**
- **The issue:** *Students are too noisy and talk loudly in class*

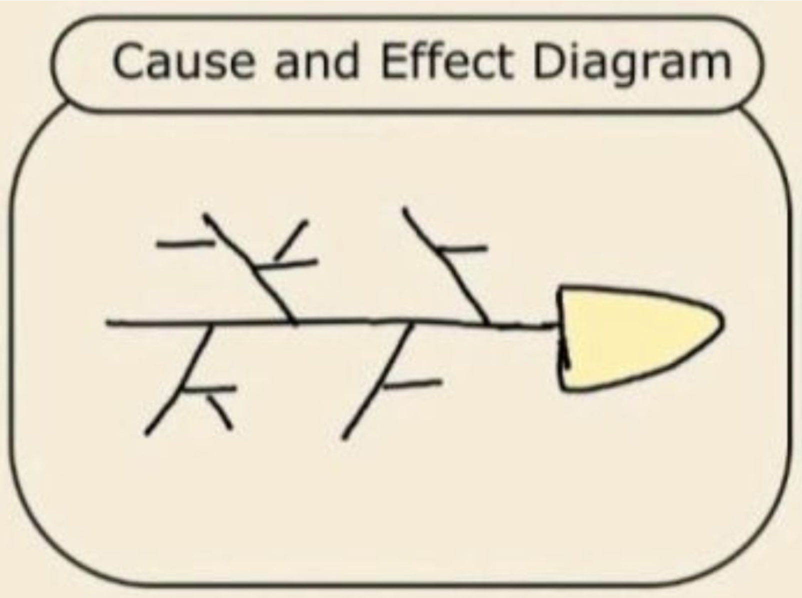


Note:

Categories when brainstorming: People, Process, Materials, Equipment, Environment

Fishbone Diagram: **Exercise 3**

- Draw a fishbone diagram to identify the **root causes** and **suggest fixes**
- **The issue:** *US places very high tariffs on Vietnam’s exports*



Note:

Categories when brainstorming: People, Process, Materials, Equipment, Environment

Summary

Summary

📌 You have learned:

- **Quality** ensures project success and stakeholder satisfaction
- The three processes – **Plan, Assure, Control** – maintain quality
- Various **tools and techniques** help analyze and improve project quality
- Quality management **models** and **certifications** drive continuous improvement

Thank you for listening

Q & A

Appendix

Project Quality Management Processes

1. **Plan Quality Management:** Setting Standards
2. **Perform Quality Assurance (QA):** Ensuring Compliance
3. **Perform Quality Control (QC):** Monitoring & Inspecting

Step 1: Plan Quality Management: **Key Activities**

1. Identify Quality Standards & Requirements

- Internal and external quality **standards (ISO, Six Sigma, PMBOK)**
- Stakeholder **expectations and regulatory requirements**

Example:

A construction project must meet safety and building code standards

Step 1: Plan Quality Management: **Key Activities**

2. Develop a Quality Management Plan

- Defines **quality metrics (e.g., defect rate, uptime, etc.)**
- Establishes **roles and responsibilities in QA and QC**

Example:

Software projects use code review metrics to ensure maintainability

Step 1: Plan Quality Management: **Key Activities**

3. Define Quality Tools & Techniques

- Tools like **Cause-and-Effect Diagrams, Control Charts, and Checklists**
- Methods like **Benchmarking, Cost-Benefit Analysis, and Sampling**

Example:

Using Six Sigma for process improvement

Step 2: Perform Quality Assurance

1. Process Audits & Reviews

- Ensures processes **align with quality standards**
- Example: **Software code review sessions to prevent bugs**

Step 2: Perform Quality Assurance

2. Continuous Process Improvement

- Uses **Total Quality Management (TQM) & Six Sigma** to refine processes

Example:

Toyota Production System focuses on minimizing waste (Lean Manufacturing)

Step 2: Perform Quality Assurance

3. Benchmarking & Best Practices

- Compares processes against **industry leaders** to identify gaps

Example:

Comparing project efficiency with top competitors in the same industry

Step 3: Perform Quality Control

1. Conduct Inspections & Testing

- Verifies **if deliverables meet quality requirements**

Example:

Software QC teams perform unit and system testing

Step 3: Perform Quality Control

2. Use Statistical Quality Control Tools

- **Control Charts** – Track variations in process performance
- **Pareto Analysis (80/20 Rule)** – Focus on fixing the most critical defects first

Example:

Manufacturing industries use Six Sigma control charts to monitor defects

Step 3: Perform Quality Control

3. Identify & Resolve Defects

- Root cause analysis (Fishbone Diagrams) to find **why defects occur**

Example:

Analyzing why a website crashes under heavy traffic loads